

SACS



Combine

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The Combine program is a utility program completely compatible with common solution files from the SACS suite of programs. It allows the user to Combine static and/or dynamic results of the same model from various analyses into a single solution file. The program has the ability to Combine results from dissimilar postfiles allowing results of various construction stages to be interpreted and can also be used to perform symmetric-antisymmetric analysis.

1.2 PROGRAM FEATURES

Combine reads model physical data, deflections and internal loads from SACS common solution files. Some of the main features and capabilities of the program are:

1. Combine results from two solution files into one solution file.
2. Create new load conditions by combining the results of existing basic load conditions either linearly or by the SRSS method (square root of the sum of the squares).
3. Combine dynamic modal responses by either the SRSS or Complete Quadratic (CQC) methods.
4. Combine static and earthquake analysis results.
5. Format or unformat solution files, i.e. convert UNIX or DOS binary solution file into ASCII file, or convert ASCII file into UNIX or DOS binary file.
6. Combine results of analyses performed at various phases of construction by allowing dissimilar solution files.
7. Calculate extreme wave internal loads.
8. Calculate internal loads for extreme wind.
9. Obtain analysis results for an entire structure from the results of a partial model by using symmetric and antisymmetric techniques.
10. Supports unlimited number of load cases.
11. Supports alpha numeric load case names.
12. Automation option to combine all load cases from one solution file with a single load case from the second solution file.

Note: When combining results of 2 or more basic load cases, the load cases need not be in the same solution file and the solution files may contain a different number of joints and/or members.

2 COMBINING ANALYSIS RESULTS

The following sections contain a brief discussion on specifying combination procedures for various types of analysis results. For a discussion on the theory of the combination techniques, see the Commentary in Section 4 of this manual.

The LCOND line is used to create new load cases from existing solution files. The new load case name, combination technique and allowable stress modifier are specified in columns 7-10, 14-17 and 18-24, respectively.

2.1 STATIC RESULTS

Results from static analyses can be combined by taking the algebraic sum (superposition), the sum of the absolute values, the square root of the sum of the squares (SRSS), or the largest value and adding the SRSS of the remaining values, by specifying 'LIN', 'PEAK', 'RMS' or 'PRMS', respectively, on the LCOND input line.

The name of the new load case being created is specified in columns 7-10 on the LCOND line. Each contributing basic load condition is specified on the ensuing COMP input lines along with its source* (Primary or Secondary file) and the load factor.

Note: When results of two solution files are being combined, one file is designated as the 'Primary file' and the other as the 'Secondary file' in the Combine runfile.

2.2 DYNAMIC RESULTS

In addition to the 'LIN', 'PEAK', 'RMS' and 'PRMS' methods, modal results can be combined using the Complete Quadratic Combination technique by specifying 'CQC' on the LCOND input line.

The name of the new load case being created is specified in columns 7-10 on the LCOND line. The contributing modes, the source file** (usually Primary) and the appropriate modal participation factor, should be specified on the ensuing COMP input line regardless of combination method. When specifying the 'CQC' method, the modal frequency and percent damping should also be input on the COMP line.

Note: Normally, only one solution file is used when combining modal results.

2.3 STATIC AND EARTHQUAKE RESULTS

Static analysis and seismic results can be combined assuming the earthquake axial stresses are either tensile or compressive by specifying 'PRST' or 'PRSC' respectively on the LCOND line. In either case, all other stresses are assumed to have the same sign as the corresponding static stress.

The name of the new load case being created is specified in columns 7-10 on the LCOND line. For a typical earthquake analysis, two combinations containing static plus seismic stresses, are made for each seismic basic load case, one combination with seismic axial loads assumed to be in tension and one with seismic axial loads assumed to be in compression.

The solution file containing static results should be specified as the Primary solution file. The earthquake solution file should be specified as the Secondary solution file in the Combine runfile.

2.4 RESULTS FROM DISSIMILAR MODELS

Solutions from models of different sizes (i.e. number of joints and/or members) can be combined using the Combine program. This capability allows the analysis results of structures in various construction stages to be combined for evaluation.

The Combine program will successfully merge different models if and only if the first model is the larger model and the joints/members of the second model are identical in order and properties to the initial joints/members of the first model. For example, if the second model consists of 40 joints and 30 members, then the members of the first model must start with a duplicate of the 30 members from the second model and the joints of the first model must start with a duplicate of the 40 joints from the second model. The first model may contain more members or joints than the second model, but the extra members/joints must be placed after the members/joints that are common to the two models.

2.5 SYMMETRIC-ANTISYMMETRIC ANALYSIS

Combine allows for the analysis of extremely large symmetric models utilizing symmetric and anti-symmetric boundary conditions.

2.6 EXTREME WAVE ANALYSIS RESULTS

The extreme wave analysis results are generated by the Combine program. The transfer function data must be created using Seastate, Wave Response, MORA or WAMIT and supplied in the form of common solution files.

The spectral analysis option must be designated by 'SP' in columns 14-15 on the CMBOPT line.

Each direction is initiated by the DIRECT input line. The wave direction along with the dead load case location and load case name are designated in columns 7-14, 15 and 16-19, respectively. The load case factor and the axial load option are designated in columns 20-26 and 27, respectively. For each wave direction, the load cases making up the transfer function and the wave spectra for which to determine the extreme wave stresses are specified using LCAS and WSPEC input lines.

11	COMP	SDEAD		0.9
12	LCOND	COM2	LIN	1.0
13	COMP	CBBBB		1.3
14	COMP	SDEAD		0.9
15	END			

The new automation feature allows to combine all load conditions in primary CSF file with a single load case in the secondary CSF with few input lines. This option also enables to re-factorize the load. For the example previously discussed, the COMBINE input file with automation options is given below.

2.8.1 New options in COMBINE input file

A series of the new options must be added to facilitate automation of COMBINE input file. The automation option becomes available by inputting the keyword "AUTO" to the columns 77-80 of CMBOPT card. The keyword "ALL" is added into columns 62-64 of LCOND input line. The new load combination factors must also be added to LCOND input line – columns 66-72 and 74-80 for Primary and Secondary CSFs, respectively. The initial load combination factors must be added in COMB input line after LCOND line.

The new options are summarize as follows:

- CMBOPT – columns 77-80: add the keyword "AUTO"
- LCOND – columns 62-64: add the keyword "ALL"
- LCOND – columns 66-72: input new factor for Primary CSF
- LCOND – columns 74-80: input new factor for Secondary CSF
- COMB P – columns 11-22: input initial factor for Primary CSF
- COMB S – columns 7-10: input the load case ID in Secondary CSF
- COMB S – columns 11-22: input initial factor for Secondary CSF

Note following comments on automation process in COMBINE program:

- Only ONE automatic LOCND input line is allowed. The automation only work for algebraic summation (linear).
- Both COMP P and COMP S lines should be presented after LCOND. In the case of additional COMP input lines, the program returns an error.
- The Primary CSF should not have more than 26000 load cases.

	1	2	3	4	5	6	7	8
1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	* Sample Problem - Auto Combine Input File							
2	CMBOPT						AUTO	
3	LCOND	LIN	1.0		ALL	1.30	0.90	
4	COMP P		1.0					
5	COMP SDEAD		-1.0					
6	END							

In a case of combining results of two Dynamic/Wave Response of a single model (identical model with same number of time steps), the COMBINE input is shown in this sample input file. COMBINE program reads same number of load cases from two primary and secondary CSFs and combines/factorizes the results.

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

1	1234567890123456789012345678901234567890123456789012345678901234567890
2	* Sample Problem - Automatically combine two Dynamic/Wave Response Analysis
3	CMBOPT
4	LCOND LIN 1.0 ALL AUTO
5	COMP P 1.0
6	COMP S 1.0
7	END

3 COMMENTARY

3.1 COMBINATION METHOD

The square root of the sum of the squares combination method, usually called the SRSS or RMS method, is used to Combine analysis results that are not correlated and are not direction dependent. The RMS result is taken as:

$$r_{\text{RMS}} = \sqrt{r_1^2 + r_2^2 + r_3^2 + \dots + r_n^2}$$

3.2 COMPLETE QUADRATIC COMBINATION METHOD

The application of two-dimensional methods to the three-dimensional dynamic analysis has resulted in the use of combination procedures that assume that only responses in the same direction as the load occur. The contribution of modal responses orthogonal to the load are usually not considered.

The Complete Quadratic Combination 'CQC' method on the other hand, accounts for the cross-correlation or coupling among responses of closely spaced modes. This method recognizes that when a particular mode responds, a secondary coupled response may occur from modes whose frequencies are closely spaced. The contribution of these secondary responses can be expressed as a coefficient which is a function of the modal frequencies, structural damping ratios and the frequency content of the load.

Assuming that the periods of the structural modes are short compared to the earthquake duration, and the earthquake spectrum is smooth over a wide range of frequencies, the cross-correlation coefficient between modal responses i and j , can be calculated from:

$$\rho_{ij} = \frac{8\sqrt{\zeta_i\zeta_j}(\zeta_i + r\zeta_j)r^{3/2}}{(1 - r^2)^2 + 4\zeta_i\zeta_jr(1 + r^2) + 4(\zeta_i^2 + \zeta_j^2)r^2}$$

where $r = \omega_j/\omega_i$.

For constant modal damping, $\zeta_i = \zeta_j = \text{constant}$, the cross correlation coefficient can be taken as:

$$\rho_{ij} = \frac{8\zeta^2(1 + r)r^{3/2}}{(1 - r^2)^2 + 4\zeta^2r(1 + r)^2}$$

Note: For equal damping, modes with the same frequency ($r = 1$), will respond equally regardless of direction ($\rho_{ij} = 1$). The CQC method is most applicable for symmetric structures with orthogonal modes of nearly the same frequency. For systems in which the frequencies are well separated, the CQC method solution degenerates to that of the SRSS method.

3.3 COMBINING STATIC AND SEISMIC RESULTS

For an earthquake analysis, the design forces for a member are obtained by combining gravity, buoyancy and hydrostatic induced forces with the forces resulting from earthquake ground motion. Because

earthquake induced forces are cyclic in nature and have no sense of direction associated with them, relative signs of these forces should be selected so that the most conservative condition results.

The 'PRST' and 'PRSC' combination methods are used to Combine seismic stresses with static stresses. For either method, member shear, bending and torsion stresses due to seismic ground motion are assumed to have the same sign as the corresponding static stress. The seismic axial stress is assumed to be in tension for the 'PRST' method and in compression for the 'PRSC' method. The seismic stresses are then linearly combined with the static stresses.

Note: Normally, this results in the creation of two static plus seismic load combinations to be used for code check purposes, for each seismic load case.

3.4 SPECTRAL RESPONSE ANALYSIS

Spectral response analysis is used to account for the random nature of a confused sea in a rational manner. The method assumes that there is a definable relation between wave height and structural response, and that at any point the elevation of the sea above its mean value is a stationary Gaussian random process.

3.4.1 Linear Systems

$$Y(f) = H(f)X(f) \quad (1)$$

It is shown in standard references that linear systems whose properties do not change with time can be characterized in the frequency domain by an expression of the form:

$$Y_i(f) = H_i(f)X(f)$$

where: f = Frequency.

$X(f)$ = Fourier transform of the excitation.

$Y(f)$ = Fourier transform of the response.

$H(f)$ = Transfer function.

The transfer function (also called the frequency response function) can be thought of as the amplitude of the sinusoidal response when the excitation is a sinusoid of unit amplitude. Equation (1) can be extended to the case of many response functions to a given excitation by interpreting the terms in a matrix sense. In subscripted notation it is written as:

$$Y_i(f) = H_i(f)X(f) \quad (2)$$

In equation (2) Y and H are $N \times 1$ matrices (or N component vectors) and X is a scalar (or a 1×1 matrix). Taking the outer product of equation (2) with itself results in the following:

$$Y_i(f)Y_j(f) = H_i(f)H_j(f)X^2(f) \quad (3)$$

If the excitation, $x(t)$, is a random function of time, then its Fourier transform, $X(f)$, is also a random function, as are those of the responses, $Y_i(f)$. In this case equation (3) is a relation between random functions (note, however, that the transfer functions, $H_i(f)$, are well defined and not random).

The average value of a random variable, Z , is represented by the notation $\underline{Z}$. The average of both sides of equation (3) gives:

$$\underline{Y_i Y_j} = H_i H_j \underline{X^2} \quad (4)$$

Including only the diagonal terms of this matrix equation yields:

$$\underline{Y_i^2} = H_i^2 \underline{X^2} \quad (5)$$

For any random function defined in the frequency domain, $Z(f)$, the function $Z^2(f)$ is called the power spectral density (or the mean-square spectral density) of the process and is designated by:

$$S_Z(f) = \underline{Z^2(f)} \quad (6)$$

The mean-square value of a stationary random function of time, $y(t)$, (i.e. a process whose statistics do not change with time) is given by:

$$\underline{y^2(t)} = \int_0^\infty \underline{Y_i^2(f)} df \quad (7)$$

The square root of this is called the root-mean-square (RMS) value. Combining this definition with equations (5), (6) and (7) yields the RMS value of the response of the system:

$$Y_{\text{RMS}} = \sqrt{\int_0^\infty H_i^2(f) S_X(f) df} \quad (8)$$

For spectral response analysis of offshore structures, the excitation is the elevation of the water surface at a point as a function of time, $h(t)$, and the responses of interest. The response is defined as the difference between successive maximum and minimum peaks in the plot of versus time.

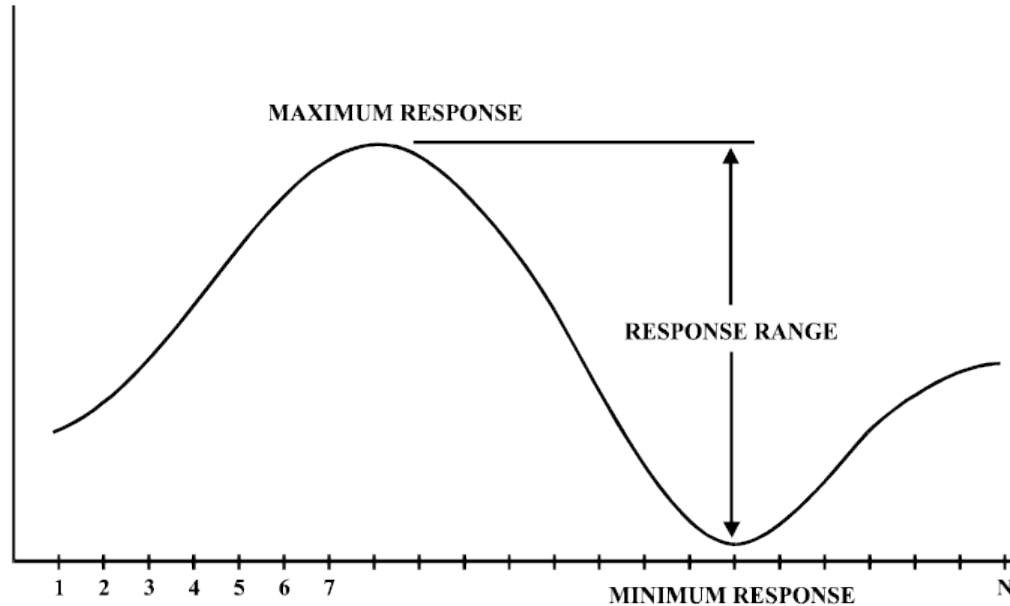
Thus if the spectral density of a particular seastate $S_h(f)$, is known, and the transfer function $H_i(f)$ can be calculated, then the statistical RMS response for this particular sea state may be taken as:

$$\sigma_{\text{RMS}} = \sqrt{\int_0^\infty H_i^2(f) S_h(f) df} \quad (9)$$

3.4.2 Transfer Function

A transfer function defines the ratio of the cyclic response to wave height as a function of frequency for a particular wave direction. If, for each frequency, the input to the system is a unit amplitude sinusoid of that frequency, then the steady state amplitude of the response is the transfer function at that frequency.

To generate a transfer function for a particular load case or wave direction, several waves of various frequencies (periods) are used to load the structure. These loads can be time history loads or real and imaginary components. The response of the structure is calculated and the difference between the maximum and minimum response is determined for each wave.



Dividing these response ranges by one-half of the corresponding wave height produces responses for waves of unit amplitude because wave height equals twice the wave amplitude for sinusoidal waves. The relationship between the response ranges of unit amplitude and the corresponding wave frequency for all waves considered is the transfer function.

3.4.3 Wave Height Spectral Density

Wave height spectra are used to characterize the random behavior of waves statistically. From a wave spectrum, a wave height spectral density relating the probability distribution at various frequencies can be developed. Three forms of wave height spectral density functions are commonly used in the offshore industry, all of which are incorporated into the Combine program; they are:

A. PIERSON-MOSKOWITZ SPECTRUM (BRETSCHNEIDER'S FORM)

$$S_{PM}(F^*) = \frac{5h_s^2 T_0}{16} \frac{1}{(F^*)^5} \exp \left[-\frac{5}{4} (F^*)^{-4} \right]$$

B. JONSWAP (JOINT NORTH SEA WAVE PROJECT) SPECTRUM

$$S_J(F^*) = \frac{S_{PM}(F^*)}{C} \exp \left\{ \ln \gamma \exp \left[-\frac{(F^* - 1)^2}{2\sigma^2} \right] \right\}$$

where:

h_s = Significant wave height, defined as the average height of the 1/3 highest waves.

T_0 = Dominant wave period, the period for which $S(f)$ is a maximum.

F^* = Dimensionless frequency, f/f_0 , where f_0 is the frequency corresponding to T_0 .

λ , σ , and C are parameters characterizing the JONSWAP spectrum. The following defaults are built into the program:

$\gamma = 3.3$

$C = 1.525$

σ is 0.07 for $F^* < 1$ and 0.09 for $F^* > 1$

C. OCHI-HUBBLE DOUBLE PEAK SPECTRUM

$$S_{OH}(f) = \frac{\pi}{2} \sum_{j=1}^2 \frac{[4(4\lambda_j + 1)\pi^4 f_{pj}^4]^{\lambda_j}}{\Gamma(\lambda_j)} \frac{h_{sj}^2}{(2\pi f)^{4\lambda_j+1}} \exp \left[-\frac{4\lambda_j + 1}{4} \left(\frac{f_{pj}}{f} \right)^4 \right]$$

where:

f = wave frequency

λ = peakedness

h_s = significant wave height

f_p = spectral peak frequency

3.4.4 Spectral Response

The RMS stress for a particular wave spectrum can be calculated from the following:

$$\sigma_{RMS} = \sqrt{\int_0^{\infty} H^2(f) S_i(f) df}$$

$S_i(f)$ is the wave height spectral density and $H(f)$ is the transfer function for the direction being considered.

If a linear system is excited by a Gaussian random process, then the response will also be a Gaussian process; thus having assumed system linearity and Gaussian excitation, the stress time histories are Gaussian at least to the order of our approximations. Further, if each response is assumed to be narrow banded, then the spectral density of the response is significant only over a narrow range of frequencies. Under these conditions the stress range is a Rayleigh distributed random variable having a probability density function given by:

$$p(s) = \frac{s}{\sigma_{\text{RMS}}^2} \exp \left[-\frac{s^2}{2\sigma_{\text{RMS}}^2} \right]$$

where s is the response range and σ_{RMS} is the RMS value of the response range.

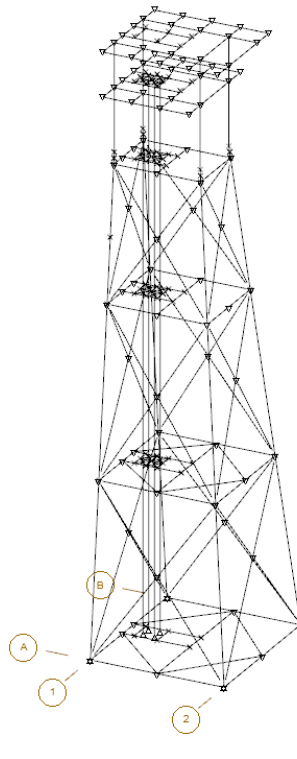
The maximum response may be calculated using the following:

$$\sigma_{\text{max}} = \sqrt{-2 \ln(1 - 0.01 \text{ PROB})} \times \sigma_{\text{RMS}}$$

4 SAMPLE PROBLEMS

The structure shown below was used to demonstrate the various capabilities of the Combine program. Three separate Combine operations are illustrated:

1. The first sample problem illustrates the ability of the Combine program to create one solution file containing results from two separate solution files.
2. Sample Problem 2 illustrates the program's ability to Combine modal results using the Complete Quadratic Combination technique.
3. Sample Problem 3 illustrates the program's ability to Combine earthquake and static results per API-RP2A guidelines.



4.1 SAMPLE PROBLEM 1 – ORIGINAL COMBINE – DIFFERENT SUPPORT CONDITIONS

Two analyses were run for the same model with different support conditions. The results were combined into one solution file so that redesign could be done for the overall critical condition.

The first solution file was designated as the primary solution file and contained results for two load cases. The second solution file contained results for one load case and was designated as the secondary solution file.

Below is the Combine input file for this sample problem followed by an explanation of the input lines used.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	* ORIGINAL COMBINE - DIFFERENT SUPPORT CONDITIONS							
2	CMBOPT							
3	LCOND	AAAA	LIN		1.0			
4	COMP	PCMB1			1.0			
5	COMP	SDEAD			-1.0			
6	LCOND	BBBB	LIN		1.0			
7	COMP	PCMB2			1.0			
8	COMP	SDEAD			-1.0			
9	LCOND	COM1	LIN		1.0			
10	COMP	CAAAA			1.3			
11	COMP	SDEAD			0.9			
12	LCOND	COM2	LIN		1.0			
13	COMP	CBBBB			1.3			
14	COMP	SDEAD			0.9			
15	END							

1. The first line is a title line.
2. The CMBOPT is left blank.
3. The first LCOND line specifies that, by default, a linear combination is to be performed when creating load case AAA (columns 14-17 left blank).
4. The COMP line specifies the following:
 - a. Load case one in the new solution file will consist of load case CMB1 of the primary solution file ('P' in column 6 and 'CMB1' in column 10) combined to the load case from secondary solution file.
 - b. The load case factor is 1.0 as designated by '1.0' in columns 12-14.
5. The COMP line specifies the following:
 - a. Load case one in the new solution file will consist of load case DEAD of the secondary solution file ('S' in column 6 and 'DEAD' in column 10) combined to the load case from primary solution file.
 - b. The load case factor is -1.0 as designated by '1.0' in columns 12-14.

6. The next LCOND line specifies that, by default, a linear combination is to be performed when creating load case BBB (columns 14-17 left blank).

7. The COMP line specifies the following:

- a. Load case two in the new solution file will consist of load case CMB2 of the primary solution file ('P' in column 6 and 'CMB2' in column 10) combined to the load case from secondary solution file.
- b. The load case factor is 1.0 as designated by '1.0' in columns 12-14.

8. The COMP line specifies the following:

- a. Load case one in the new solution file will consist of load case DEAD of the secondary solution file ('S' in column 6 and 'DEAD' in column 10) combined to the load case from primary solution file.
- b. The load case factor is --1.0 as designated by '1.0' in columns 12-14.

9. The next LCOND line specifies that, by default, a linear combination is to be performed when creating load case COM1 (columns 14-17 left blank).

10. The COMP line specifies the following:

- a. Load case two in the new solution file will consist of load case AAA of the primary solution file ('P' in column 6 and 'AAA' in column 10) combined to the load case from secondary solution file.
- b. The load case factor is 1.3 as designated by '1.3' in columns 12-14.

11. The COMP line specifies the following:

- a. Load case one in the new solution file will consist of load case DEAD of the secondary solution file ('S' in column 6 and 'DEAD' in column 10) combined to the load case from primary solution file.
- b. The load case factor is -0.9 as designated by '-0.9' in columns 12-14.

12. The next LCOND line specifies that, by default, a linear combination is to be performed when creating load case COM2 (columns 14-17 left blank).

13. The COMP line specifies the following:

- a. Load case two in the new solution file will consist of load case BBB of the primary solution file ('P' in column 6 and 'BBB' in column 10) combined to the load case from secondary solution file.
- b. The load case factor is 1.3 as designated by '1.3' in columns 12-14.

14. The COMP line specifies the following:

a. Load case one in the new solution file will consist of load case DEAD of the secondary solution file ('S' in column 6 and 'DEAD' in column 10) combined to the load case from primary solution file.

b. The load case factor is -0.9 as designated by '-0.9' in columns 12-14.

The following page contains the output listing for this sample problem.

***** COMBINE PROGRAM OPTIONS *****									
LOAD CASE	NUMB	COMB	AMOD	LOAD CASE DESCRIPTION		UNIT ORIG	FACTOR	SIGN	CHANGE
STRES	COMB	TYPE	FACT			LC =		ROT	DEFL
								XYZ	XYZ
AAAA		LIN	1.000						
	1					P CMB1	1.000000	000	000 0
	2					S DEAD	-1.000000	000	000 0
BBBB		LIN	1.000						
	1					P CMB2	1.000000	000	000 0
	2					S DEAD	-1.000000	000	000 0
COM1		LIN	1.000						
	1					C AAAA	1.300000	000	000 0
	2					S DEAD	0.9000000	000	000 0
COM2		LIN	1.000						
	1					C BBBB	1.300000	000	000 0
	2					S DEAD	0.9000000	000	000 0
FORCES AND MOMENTS SUMMARY									
*** MOMENTS SUMMED ABOUT ORIGIN ***									
		***** KIPS *****		***** FT-KIPS *****					
LOAD CASE		FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)		
AAAA		-99.930	-0.311	1131.280	493.294	1487.942	-4.371		
BBBB		-70.676	-66.751	1131.905	-4483.114	-576.442	-307.133		
COM1		-129.909	-0.404	2392.222	635.638	-3660.542	-5.681		
COM2		-91.879	-86.776	2393.035	-5833.713	-6344.247	-399.272		

4.2 SAMPLE PROBLEM 2 – AUTO COMBINE – DIFFERENT SUPPORT CONDITIONS

Two analyses were run for the same model with different support conditions. The results were combined into one solution file so that redesign could be done for the overall critical condition.

The first solution file was designated as the primary solution file and contained results for two load cases. The second solution file contained results for one load case and was designated as the secondary solution file.

Below is the Combine input file for this sample problem followed by an explanation of the input lines used.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	* AUTO COMBINE - DIFFERENT SUPPORT CONDITIONS							
2	CMBOPT							
3	LCOND COM1	LIN				ALL	1.3	AUTO 0.9
4	COMP P		1.0					
5	COMP SDEAD		-1.0					
6	END							

1. The first line is a title line.
2. The CMBOPT specifies the following:
 - a. The Auto-Combine feature is designated by 'AUTO' in columns 77-80.
3. The first LCOND line specifies the following:
 - a. A linear combination is to be performed when creating load cases (columns 14-17 set as 'LIN').
 - b. The program automatically combine all load cases (columns 62-64 set as 'ALL').
 - c. New factor for the secondary file (columns 66-72).
 - d. New factor for the primary file (74-80).
4. The COMP line specifies the following:
 - a. The load cases considered are the ones in the primary common solution file. ('P' in column 6 and '1' in column 10).
 - b. All load cases are going to be used to combine (columns 7-10 left black).
 - c. The load case factor is 1.0 as designated by '1.0' in columns 12-14.
5. The ensuing COMP line specifies the following:
 - a. The load cases considered are the ones in the secondary common solution file. ('S' in column 6 and '1' in column 10).
 - b. Dead Load Case is going to be used to combine (columns 7-10 set as 'DEAD').

c. The load case factor is -1.0 as designated by '1.0' in columns 12-14.

The following page contains the output listing for this sample problem.

***** COMBINE PROGRAM OPTIONS *****										
LOAD CASE	NUMB COMB	COMB TYPE	AMOD FACT	LOAD CASE DESCRIPTION		UNIT ORIG LC =	FACTOR	SIGN ROT	CHANGE DEFL	
STRES								XYZ	XYZ	
A000	1	LIN	1.000	1.000 * DEAD -	1.000 * DEAD	P DEAD	1.000000	000	000	0
	2					S DEAD	-1.000000	000	000	0
A001	1	LIN	1.000	1.000 * CMB1 -	1.000 * DEAD	P CMB1	1.000000	000	000	0
	2					S DEAD	-1.000000	000	000	0
A002	1	LIN	1.000	1.000 * CMB2 -	1.000 * DEAD	P CMB2	1.000000	000	000	0
	2					S DEAD	-1.000000	000	000	0
000A	1	LIN	1.000	1.300 * A000 +	0.900 * DEAD	C A000	1.300000	000	000	0
	2					S DEAD	0.900000	000	000	0
001A	1	LIN	1.000	1.300 * A001 +	0.900 * DEAD	C A001	1.300000	000	000	0
	2					S DEAD	0.900000	000	000	0
002A	1	LIN	1.000	1.300 * A002 +	0.900 * DEAD	C A002	1.300000	000	000	0
	2					S DEAD	0.900000	000	000	0
FORCES AND MOMENTS SUMMARY										
*** MOMENTS SUMMED ABOUT ORIGIN ***										
LOAD CASE	FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)				
A000	0.000	0.000	0.000	-0.001	0.004	0.000				
A001	-99.930	-0.311	1131.280	493.294	1487.942	-4.371				
A002	-70.676	-66.751	1131.905	-4483.114	-576.442	-307.133				
000A	0.000	0.000	921.559	-5.636	-5594.866	-0.001				
001A	-129.909	-0.404	2392.222	635.638	-3660.542	-5.681				
002A	-91.879	-86.776	2393.035	-5833.713	-6344.247	-399.272				

5 INPUT LINES

COMBINE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE CONTROLS THE INPUT AND OUTPUT TYPE FOR THIS ANALYSIS.
(1- 6)	ENTER 'CMBOPT' ON THIS LINE. NO HEADER LINE IS REQUIRED.
(8- 9)	ENTER 'FI' IF THE INPUT POSTFILE(S) IS FORMATTED. NORMALLY ALL POSTFILES ARE UNFORMATTED, BUT THE OPTION TO FORMAT POSTFILES ENABLES THE USER TO TRANSFER THE FILES FROM ONE COMPUTER TO ANOTHER.
(11-12)	ENTER 'FO' IF THE OUTPUT FILE IS TO FORMATTED.
(14-15)	ENTER 'SP' IF THIS ANALYSIS IS A SPECTRAL COMBINATION.
(17-18)	ENTER UNITS OPTION FOR SPECTRAL COMBINATION: 'EN' - ENGLISH. 'MN' - METRIC WITH KILONETONS. 'ME' - METRIC WITH KILOGRAMS.
(20-21)	ENTER 'JA' IF JOINT ACCELERATIONS ARE TO BE PRINTED. THIS OPTION IS ONLY VALID WITH THE SPECTRAL OPTION.
(37-42)	FOR WIND SPECTRAL ANALYSIS, ENTER THE MEAN WIND SPEED AVERAGING TIME. THIS VALUE IS NORMALLY 600 SECONDS.
(77-80)	ENTER 'AUTO' TO PERFORM AUTOMATIC COMBINE OF ALL LOAD

LINE LABEL	FORMATTED INPUT FILE OPTION	FORMATTED OUTPUT FILE OPTION	SPECTRAL ANALYSIS OPTION	SPECTRAL UNITS OPTION	JOINT ACCEL OPTION	MEAN WIND AVERAGING TIME	LEAVE BLANK	Is an auto-combine run?
CMBOPT								
1-- 6	8-- 9	11--12	14--15	17--18	20--21	37<--42	43-----76	77--80
DEFAULT						600		
ENGLISH						SEC		
METRIC						SEC		

DIRECTION HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE INDICATES THAT A NEW DIRECTION IS BEGINNING FOR SPECTRAL COMBINATION. FOLLOWING THIS LINE, THE TRANSFER FUNCTION FOR THIS DIRECTION IS DEFINED BY A SERIES OF 'LCAS' LINES WHICH ARE THEN IN TURN FOLLOWED BY THE 'WSPEC' LINES FOR THIS DIRECTION. A SIMILAR SET OF DATA IS INPUT FOR EACH DIRECTION DESIRED.
(1- 6)	ENTER 'DIRECT' ON THIS LINE. NO HEADER LINE IS REQUIRED.
(7-14)	ENTER THE DIRECTION FOR THIS SET OF DATA. THIS IS USED ONLY FOR TITLING PURPOSES.
(15)	ENTER THE SOURCE OF THE DEAD LOAD CASE: 'P' - PRIMARY FILE. 'S' - SECONDARY FILE. 'C' - PREVIOUS COMBINATION.
(16-19)	ENTER THE DEAD LOAD CASE NAME.
(20-26)	ENTER THE FACTOR FOR THE DEAD LOAD CASE.
(27)	SELECT THE TYPE OF AXIAL LOAD TO BE USED IN COMBINING WITH THE DEAD LOAD CASE: 'C' - COMPRESSION. 'T' - TENSION. 'B' - BOTH (CREATES 2 LOAD CASES FOR EACH SPECTRUM).

LINE LABEL	DIRECTION	DEAD LOAD CASE				REMARKS
		LOAD CASE SOURCE	LOAD CASE NAME	LOAD CASE FACTOR	AXIAL LOAD OPTION	
DIRECT						
1-- 6	7<--14	15	16--19	20<--26	27	28-----80
DEFAULT		'P'		1	'B'	
ENGLISH	DEG					
METRIC	DEG					

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS THE LAST LINE OF THE INPUT FILE.
(1- 3)	ENTER 'END'.

LINE LABEL	LEAVE THIS FIELD BLANK
END	
1-- 3	4-----80

COMMENTARY

```

GENERAL      THIS LINE DESIGNATES THE CONTRIBUTION OF INPUT LOAD CASES TO
              THE TRANSFER FUNCTION. EACH 'LCAS' DATA AND CONTINUATION
              LINES DESCRIBE ONE POINT ON THE TRANSFER FUNCTION FOR A
              DIRECTION.

( 1- 4)      ENTER 'LCAS' ON THIS LINE. NO HEADER LINE IS REQUIRED.

( 6-12)      ENTER THE WAVE HEIGHT THAT WAS USED TO CREATE THIS TRANSFER
              FUNCTION POINT.

(13-19)      ENTER THE WAVE PERIOD FOR THIS TRANSFER FUNCTION POINT. THE
              TRANSFER FUNCTION MUST BE DESCRIBED IN ORDER OF DESCENDING
              WAVE PERIODS.

(20-22)      ENTER THE TYPE OF LOAD COMBINATION IS TO BE USED IN CREATING
              THIS TRANSFER FUNCTION POINT. THE CHOICES ARE:

              'STD' - LINEAR LOAD CASE ADDITIONS.

              'R+I' - REAL AND IMAGINARY LOAD CASES (REQUIRES 2 LOAD CASES).

              'SIN' - SINUSOIDAL LOAD CASES (REQUIRES 2 LOAD CASES).

              'SRC' - SEARCH (FINDS MAX. AND MIN. FROM SPECIFIED LOAD
                      CASES).

( 23 )      ENTER THE SOURCE OF THIS LOAD CASE:

              'P' - PRIMARY FILE.

              'S' - SECONDARY FILE.

              'C' - PREVIOUS COMBINATION.

(24-27)      ENTER THE LOAD CASE NAME FOR THIS CONTRIBUTION.

(28-34)      ENTER THE FACTOR FOR THIS LOAD CASE.

(35-46)      ENTER DATA FOR 2ND LOAD CASE IF NEEDED.

(47-58)      ENTER DATA FOR 3RD LOAD CASE IF NEEDED.

(59-70)      ENTER DATA FOR 4TH LOAD CASE IF NEEDED.

              REPEAT THIS LINE FOR ADDITIONAL LOAD CASES IF NEEDED. THE
              WAVE HEIGHT, WAVE PERIOD, AND COMBINATION TYPE FIELDS MUST BE
              LEFT BLANK FOR THE ADDITIONAL LINES.

              REPEAT THIS LINE FOR ALL TRANSFER FUNCTION POINTS FOR THIS
              DIRECTION.

```

[illegible]

USER-DEFINED SPECTRUM

COLUMNS	COMMENTARY
GENERAL	THIS LINE DEFINES THE USER SPECTRUM. FOR THOSE CASES WHERE A PIERSON-MOSKOWITZ OR JONSWAP SPECTRUM DEFINITION IS NOT SATISFACTORY, THE USER CAN DEFINE A PARTICULAR SPECTRUM WITH UP TO 100 POINTS USING 4 VALUES PER RECORD AND UP TO 25 RECORDS. THESE RECORDS FOLLOW THE 'WSPEC' LINE WITH THE 'US' OPTION.
(1- 4)	ENTER 'SPEC'.
(9-16)	ENTER THE PERIOD OF THE FIRST POINT OF THE WAVE SPECTRUM. PERIODS MUST BE ENTERED IN DESCENDING ORDER. A PERIOD OF ZERO SHOULD NEVER BE ENTERED.
(17-24)	ENTER THE WAVE SPECTRUM VALUE IN TERMS OF WAVE HEIGHT SQUARED OVER HERTZ.
(25-40)	ENTER THE SECOND POINT.
(41-56)	ENTER THE THIRD POINT.
(57-72)	ENTER THE FOURTH POINT.

LINE LABEL	USER DEFINED SPECTRAL DATA								LEAVE BLANK
	1ST POINT		2ND POINT		3RD POINT		4TH POINT		
	PERIOD	SPECTRA VALUE	PERIOD	SPECTRA VALUE	PERIOD	SPECTRA VALUE	PERIOD	SPECTRA VALUE	
SPEC									
1-- 4	9<--16	17<--24	25<--32	33<--40	41<--48	49<--56	57<--64	65<--72	73--80
DEFAULT									
ENGLISH	SEC	FT**2/HZ	SEC	FT**2/HZ	SEC	FT**2/HZ	SEC	FT**2/HZ	
METRIC	SEC	M**2/HZ	SEC	M**2/HZ	SEC	M**2/HZ	SEC	M**2/HZ	

DESCRIPTIVE TITLE

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	ALL TITLE LINES ARE ALWAYS PLACED IMMEDIATELY BEFORE THE 'CMBOPT' LINE.
GENERAL	THE USER CAN PLACE AS MANY DESCRIPTIVE TITLE LINES AS REQUIRED IN THE INPUT FILE. THE FIRST TITLE LINE IS PRINTED AS A HEADING AT THE TOP OF EACH OUTPUT PAGE. ALL TITLE LINES ARE LISTED AT THE BEGINNING OF THE PRINTED OUTPUT.

DESCRIPTIVE TITLE INFORMATION

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WAVE SPECTRAL DENSITY SPECIFICATION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED IF AND ONLY IF A SPECTRAL ANALYSIS IS BEING DONE. IT IS USED TO DESIGNATE THE FORM OF THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION.
(1- 5)	ENTER 'WSPEC'.
(11-12)	ENTER THE TYPE OF SPECTRUM TO BE USED FOR THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION. CHOOSE FROM BETWEEN THE FOLLOWING: 'PM' - PIERSON-MOSKOWITZ SPECTRUM. THIS IS THE DEFAULT. 'JS' - JONSWAP SPECTRUM. 'US' - USER-DEFINED SPECTRUM.
(13-19)	ENTER THE 'SIGNIFICANT WAVE HEIGHT' FOR THIS SPECTRUM. FOR USER-DEFINED SPECTRUM, THIS VALUE SQUARED WILL BE USED TO MULTIPLY THE INPUT SPECTRUM. ALSO FOR USER-DEFINED SPECTRUM, THIS VALUE DEFAULTS TO 1.0.
(20-26)	ENTER THE "DOMINANT PERIOD" FOR THIS SPECTRUM. FOR USER-DEFINED SPECTRUM, THIS VALUE WILL BE USED TO MULTIPLY THE INPUT SPECTRUM. ALSO FOR USER-DEFINED SPECTRUM, THIS VALUE DEFAULTS TO 1.0.
(34-47)	ENTER THE VALUES OF THE PARAMETERS "GAMMA" AND "C" REQUIRED TO FULLY DEFINE THE JONSWAP SPECTRUM IF 'JS' IS IN COLUMNS 11-12.
(48-56)	ENTER THE PROBABILITY LEVEL OF EXCEEDANCE. THIS IS THE PROBABILITY THAT THE QUANTITY CALCULATED WILL NOT BE EXCEEDED BASED ON A RAYLEIGH PROBABILITY DISTRIBUTION.
(57-58)	THE RESULTS GENERATED BY THE COMBINE PROGRAM CAN EITHER BE SINGLE AMPLITUDE ('SA') OR DOUBLE AMPLITUDE ('DA'). THE SPECTRUM ENTERED IS ASSUMED DOUBLE AMPLITUDE. IN MOST CASES, FOR AN EXTREME WAVE ANALYSIS SINGLE AMPLITUDE RESULTS ARE DESIRED.

LINE LABEL	WAVE SPECTRUM TYPE	WAVE SPECTRUM PARAMETERS		JONSWAP PARAMETERS		PROBABILITY LEVEL OF EXCEEDANCE	SINGLE OR DOUBLE AMPLITUDE OPTION	LEAVE THIS FIELD BLANK
		SIGNIFICANT WAVE HEIGHT	DOMINANT PERIOD	"GAMMA"	"C"			
WSPEC								
1-- 5	11--12	13<--19	20<--26	34<--40	41<--47	48<--56	57--58	59--80
DEFAULT	'PM'			3.3	1.525	99.9	'SA'	
ENGLISH		FT	SEC			PERCENT		
METRIC		M	SEC			PERCENT		

COMBINE OPTIONS LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE CONTROLS THE INPUT AND OUTPUT TYPE FOR THIS ANALYSIS.
(1- 6)	ENTER 'CMBOPT' ON THIS LINE. NO HEADER LINE IS REQUIRED.
(8- 9)	ENTER 'FI' IF THE INPUT POSTFILE(S) IS FORMATTED. NORMALLY ALL POSTFILES ARE UNFORMATTED, BUT THE OPTION TO FORMAT POSTFILES ENABLES THE USER TO TRANSFER THE FILES FROM ONE COMPUTER TO ANOTHER.
(11-12)	ENTER 'FO' IF THE OUTPUT FILE IS TO FORMATTED.
(14-15)	ENTER 'SP' IF THIS ANALYSIS IS A SPECTRAL COMBINATION.
(17-18)	ENTER UNITS OPTION FOR SPECTRAL COMBINATION: 'EN' - ENGLISH. 'MN' - METRIC WITH KILONETONS. 'ME' - METRIC WITH KILOGRAMS.
(20-21)	ENTER 'JA' IF JOINT ACCELERATIONS ARE TO BE PRINTED. THIS OPTION IS ONLY VALID WITH THE SPECTRAL OPTION.
(37-42)	FOR WIND SPECTRAL ANALYSIS, ENTER THE MEAN WIND SPEED AVERAGING TIME. THIS VALUE IS NORMALLY 600 SECONDS.
(77-80)	ENTER 'AUTO' TO PERFORM AUTOMATIC COMBINE OF ALL LOAD

LINE LABEL	FORMATTED INPUT FILE OPTION	FORMATTED OUTPUT FILE OPTION	SPECTRAL ANALYSIS OPTION	SPECTRAL UNITS OPTION	JOINT ACCEL OPTION	MEAN WIND AVERAGING TIME	LEAVE BLANK	Is an auto-combine run?
CMBOPT								
1-- 6	8-- 9	11--12	14--15	17--18	20--21	37<--42	43-----76	77--80
DEFAULT						600		
ENGLISH						SEC		
METRIC						SEC		

LOAD CONDITION COMPONENTS

COLUMNS	COMMENTARY
GENERAL	THIS LINE DESIGNATES THE CONTRIBUTION OF AN INPUT LOAD CASE.
(1- 4)	ENTER 'COMP' ON THIS LINE. NO HEADER LINE IS REQUIRED.
(5- 6)	ENTER THE LOAD CONDITION SOURCE: ' P' - PRIMARY FILE (FILE NUMBER 1). ' S' - SECONDARY FILE (FILE NUMBER 2). ' C' - PREVIOUS COMBINATION. '##' - ENTER FILE NUMBER 3-99. REPEAT THIS LINE FOR EACH LOAD CASE CONTRIBUTION.
(7-10)	ENTER THE NAME OF THIS CONTRIBUTING LOAD CASE.
(11-22)	ENTER THE FACTOR TO MULTIPLY THIS LOAD CASE. DEFAULT IS 1.0.
(25-31)	IF HALF OF A SYMMETRIC STRUCTURE HAS BEEN MODELED, THESE SIGN CHANGES CAN BE USED TO SIMULATE THE OMITTED HALF OF THE STRUCTURE.
(25)	ENTER '1' TO CHANGE SIGN ON ROTATIONS AND MOMENTS ABOUT THE X AXIS.
(26)	ENTER '1' TO CHANGE SIGN ON ROTATIONS AND MOMENTS ABOUT THE Y AXIS.
(27)	ENTER '1' TO CHANGE SIGN ON ROTATIONS AND MOMENTS ABOUT THE Z AXIS.
(28)	ENTER '1' TO CHANGE SIGN ON DEFLECTIONS AND FORCES IN THE X DIRECTION.
(29)	ENTER '1' TO CHANGE SIGN ON DEFLECTIONS AND FORCES IN THE Y DIRECTION.
(30)	ENTER '1' TO CHANGE SIGN ON DEFLECTIONS AND FORCES IN THE Z DIRECTION.
(31)	ENTER '1' TO CHANGE SIGN ON ALL STRESSES.
(32-55)	FOR 'CQC' COMBINATIONS OR WIND SPECTRAL ANALYSIS (WSP), ENTER THE MODAL FREQUENCY AND DAMPING RATIO FOR THIS MODAL CONTRIBUTION.

LINE LABEL	SOURCE OF CONTRIBUTING LOAD CASE	CONTRIBUTING LOAD CONDITION	LOAD CASE FACTOR	SYMMETRIC-ANTISYMMETRIC MODEL SIGN CHANGES							MODAL PARAMETERS		LEAVE BLANK
				ROT. MOM. X AXIS	ROT. MOM. Y AXIS	ROT. MOM. Z AXIS	DEFL. FORCE X AXIS	DEFL. FORCE Y AXIS	DEFL. FORCE Z AXIS	ALL STRESSES	FREQUENCY (HZ)	DAMPING RATIO (PERCENT)	
COMP													
1-- 4	5-- 6	7--10	11--22	25	26	27	28	29	30	31	32--43	44--55	56--80

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS THE LAST LINE OF THE INPUT FILE.
(1- 3)	ENTER 'END'.

LINE LABEL	LEAVE THIS FIELD BLANK
END	
1-- 3	4-----80

LOAD CASE CREATION INPUT

COLUMNS	COMMENTARY
GENERAL	THIS LINE CONTROLS THE CREATION OF OUTPUT LOAD CONDITIONS. EACH 'LCOND' CREATES ONE OUTPUT LOAD CASE WHICH WILL BE NAMED IN CONSECUTIVE ASCENDING ORDER. THIS DATA ALSO CONTROLS HOW THE CONTRIBUTING LOAD CASES WILL BE COMBINED TO FORM THE OUTPUT LOAD CASE.
(1- 5)	ENTER 'LCOND' ON THIS LINE. NO HEADER LINE IS REQUIRED.
(7-10)	ENTER THE LOAD CONDITION NAME. IF LEFT BLANK, THE LOAD CONDITION IS INCREMENTED FROM THE LAST LOAD CONDITION SPECIFIED IN ASCII CHARACTER ORDER.
(14-17)	ENTER THE LOAD COMBINATION TYPE FROM THE FOLLOWING SELECTIONS: 'LIN ' - ALGEBRAIC SUM (LINEAR). 'PEAK' - SUM OF ABSOLUTE VALUES. 'RMS ' - SQUARE ROOT OF THE SUM OF THE SQUARES. 'PRMS' - PEAK OF THE LARGEST PLUS THE SQUARE ROOT OF THE SUM OF THE SQUARES OF THE REST. 'PRSC' - USED IN COMBINING DEAD LOAD WITH A SPECTRAL EARTHQUAKE ANALYSIS. THE EARTHQUAKE MEMBER LOADS WILL BE COMBINED USING THE SIGN OF THE DEAD LOADS EXCEPT FOR THE AXIAL LOADS WHICH WILL BE COMPRESSIVE. 'PRST' - SAME AS 'PRSC' EXCEPT THE EARTHQUAKE MEMBER AXIAL LOADS WILL BE TENSION. 'CQC ' - COMPLETE QUADRATIC COMBINATION. 'WSP ' - WIND SPECTRAL ANALYSIS. DYNAMIC AMPLIFICATION FACTORS WILL AUTOMATICALLY BE CALCULATED.
NOTE:	WHEN SPECIFYING TYPE 'PRSC' OR 'PRST', THE STATIC ANALYSIS POST FILE SHOULD BE SPECIFIED AS THE PRIMARY FILE AND THE EARTHQUAKE POST FILE AS THE SECONDARY FILE.
(18-24)	ENTER THE ALLOWABLE STRESS MODIFIER FOR THIS LOAD CASE.
(36-80)	ENTER ANY REMARKS TO IDENTIFY THIS LOAD CASE.

LINE LABEL	LOAD CONDITION NAME	COMBINATION TYPE	AMOD	REMARKS	AUTO COMBINE	NEW FACTOR FOR SECONDARY FILE	NEW FACTOR FOR PRIMARY FILE
LCOND							
1-- 5	7--10	14--17	18<--24	36-----60	62--64	66--72	74--80
DEFAULT		'LIN '	1			1	1

DESCRIPTIVE TITLE

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	ALL TITLE LINES ARE ALWAYS PLACED IMMEDIATELY BEFORE THE 'CMBOPT' LINE.
GENERAL	THE USER CAN PLACE AS MANY DESCRIPTIVE TITLE LINES AS REQUIRED IN THE INPUT FILE. THE FIRST TITLE LINE IS PRINTED AS A HEADING AT THE TOP OF EACH OUTPUT PAGE. ALL TITLE LINES ARE LISTED AT THE BEGINNING OF THE PRINTED OUTPUT.

DESCRIPTIVE TITLE INFORMATION

2-----78